Shahjalal University of Science and Technology
4th Year 2nd Semester Final Exam (Session 2020-21)
Course Title: Human Computer Interaction.
Course Code: CSE 495, Credit: 3.0
Marks: 100; Time: 3 hours

Note: Answers all questions, Use single script, please answer the questions in order. Possession of mobile phones is strictly prohibited

Part A

1 ***Answer any 5*** 5*2

a) Why is usability important?
b) Mention Universal Design Principal.
c) What is WIMP interface?
d) What is 3 click rules?
e) What is claim and assumption in conceptualizing the design?
f) What is mind maps?
g) What are the problems of interface metaphors?
h) What is 'Wizard-of-Oz' prototyping?

2 ***Answer any 4*** 4*5

a) Imagine you're on a journey to solve a big design problem. The "Double Diamond" is your map. Can you explain the four main territories you travel through on this map to get from the problem to the perfect solution?

b) In the story of creating a great product, the user is the hero. What are the nine key chapters in this "user-centered" story that make sure we always keep the focus on our hero from start to finish?

c) Students complain that university portal is slow and difficult to navigate. Which usability aspects are being affected. Propose a solution.

d) On our computers, we have a "desktop," "folders," and a "trash can" to help us understand how to use them without a manual. What is this clever trick of using familiar, real-world ideas called, and why is it so powerful for making new technology feel easy?

c) Picture this: you've just finished a brainstorming session and have a wall full of scattered sticky notes. What is the magical technique called that helps you sort all this chaos into neat, organized groups, and how do you perform this trick?

3 ***Answer any 2*** 2*10

a) Imagine you are a user experience (UX) designer creating a new mobile app for a local cafe that allows customers to order and pay ahead.
Given this scenario, **what is prototyping, and how would you use both low-fidelity and high-fidelity prototyping to test and refine the app's design?** What are the key differences between these two types of prototypes in this context?

b) What are the primary types of interactions in user experience (UX) design, and how would you apply each to a real-world example

c) What are the core steps of the **User-Centered Design (UCD)** process? Briefly explain how a product team might apply these steps to develop a mobile banking app.

Part B

4 ***Answer any 5*** 5*2

a) What is 10 second rule in Web design?
b) What is the issue of long URL?
c) What is the potential problem of *Think a loud* approach?
d) What are the methods of gathering subjective data?
e) Why is consent important for the participants before joining experiment?
f) Define heuristic evaluation.
g) What are the methods of predicative evaluation?

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h) What is a a people centered approach?

5 ***Answer any 4*** 4*5

a) A smart speaker often misinterprets voice commands due to background noise. What challenge does this illustrate? Suggest solution to improve user experience.

b) An ecommerce website asks users to create an account before checkout, leading many to abandon their carts. Which usability principle is ignored? Suggest alternative design to reduce the drop-offs.

c) Explain the key HCI design principles for mobile and web interfaces. Discuss the challenges like screen size, navigation, touch gesture and responsiveness.

d) A software company wants to know if a new user interface reduces the time taken to complete online transactions. Describe the key procedures in experimental design, showing how they would apply to this investigation.

e) Imagine a technology company wants to understand how elderly adults interact with smart home devices.
Given this scenario, **how would an ethnographic study be used to collect data on their user experience?** What specific methods would a researcher use to gather insights that couldn't be obtained through traditional surveys or lab tests?

6 ***Answer any 2*** 2*10

a) How would you describe the five basic steps of usability testing to a new product manager? For example, if you were testing a new e-commerce website, what would you do at each step to ensure it is user-friendly and effective?

b) What is Hypothesis? What is the purpose of Null and Alternate hypothesis?

Hypothesis: There is no difference in user performance (time and error rate) and preference (5 point likert scale) when typing on two sizes (i.e., small, large) of an alphabetic, qwerty and random on-screen keyboard using a touch-based large screen, a mouse- based monitor, or a stylus-based PDA.

What are the independent variables here? What are the levels? What are the dependent variables here? What is the purpose of control variable?

c) How can a researcher use a focus group discussion (FGD) to collect and validate data, and what are some effective interview techniques for data collection?

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